EE 354/CE 361/MATH 310 -

Introduction to Probability

and Statistics

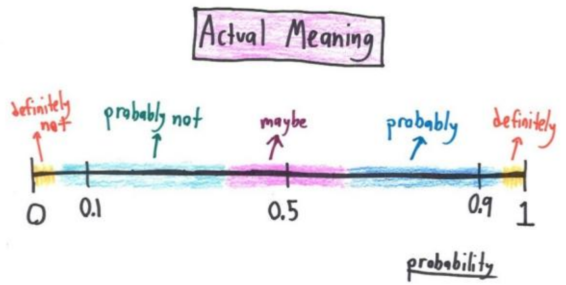
Spring 2026

EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 4 - Lecture No 08 – February 04, 2026



Announcements!

1. Office hours – Tuesday & Thursday 12:00 to 1:00 PM @ Room C-211 (or through meeting request)
2. **Quiz No 03 (Unit 3, Counting) – February 11, 2026**
3. Assignment No 01 posted – due date February 10, 2026 (11:59 PM)
4. 03 TAs included in MATH 310 team (see CANVAS course site for details)

Agenda for today

1. **Quiz No 02 (Unit 2)**
2. **Unit 3 – Counting**

TAs -

EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics, Spring 2026

Meet Your TAs

Zain Naqi

About Me

I am a Computer Science Junior. My career interests include machine learning, deep learning, and reinforcement learning. Apart from academics, I do competitive programming and run a small e-commerce business. Mostly, you will find me in the library discussion area.

Ehsaas Hours

- Tue 9:30 - 11:30
- Thu 9:30 - 11:30

Contact Me

zn09224@st.habib.edu.pk or “Zain Naqi” on Teams

Meet Your TAs

Rayyan Shah

About Me

I am a Computer Science Junior. I am interested in bioinformatics, AI, and game development. Outside of academics, I also enjoy cooking and baking, and I always have some new recipe to recommend. You can usually find me in the library or tapal.

Ehsaas Hours

- Mon 4:00-5:00
- Wed 4:00-5:00
- Fri 3:30-4:30

Contact Me

rs08770@st.habib.edu.pk

Or “Rayyan Ali Shah” on Teams

Meet Your TAs

Hiba Aiman Ali

About Me

I am a Computer Engineering Junior with a minor in South Asian Music. My interests include bioinformatics, IoT, music and sustainability. I also sing as a part of the Habib University Orchestra. When I’m not in class, I’m most likely in the music room, singing or *trying to* learn an instrument.

Ehsaas Hours

- Mon 2:30 - 4:00
- Wed 2:30 - 4:00
- Fri 12:00-1:00

Contact Me

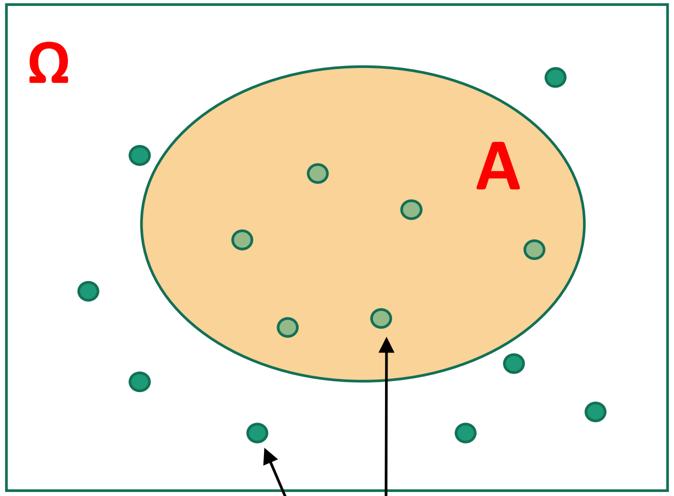
ha09269@st.habib.edu.pk on Teams or Outlook

Counting – Why?

Discrete Uniform Law:

- Assume Ω consists of n equally likely elements
- Assume A consists of k elements

Then: $$\mathbf{P}(A) = \frac{\text{number of elements of } A}{\text{number of elements of } \Omega} = \frac{k}{n}$$



$$\boldsymbol{prob = \frac{1}{n}}$$

A coin tossing problem

Assumptions:
- **independence**
- $\mathbf{P}(H) = p$

$$\mathbf{P}\,(k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$

- Given that there were 3 heads in 10 tosses, what is the probability that the first two tosses were heads?
  - event A : the first 2 tosses were heads.
  - event B : 3 out of 10 tosses were heads.

- First solution:

$$\Rightarrow \quad \mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)} = \frac{\boldsymbol{P}(\boldsymbol{H_1}\ \boldsymbol{H_2}\ \textbf{\textit{and one H in (between) tosses}}\ \mathbf{3, 4, \ldots, 10})}{\boldsymbol{P}(B)}$$

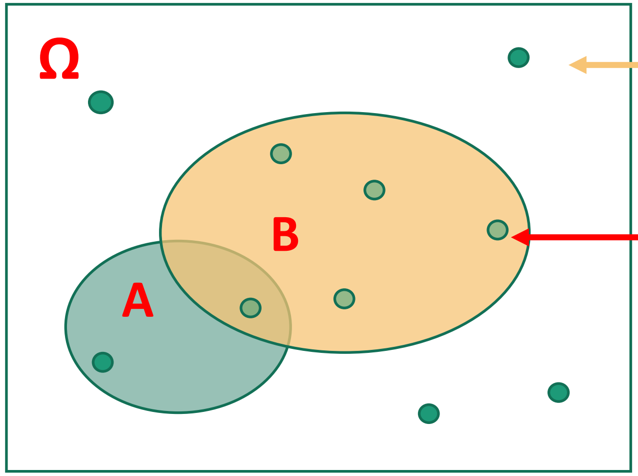
$$\Rightarrow \quad = \frac{p^2 \cdot \binom{8}{1} p^1 \cdot (1-p)^7}{\binom{10}{3} p^3 (1-p)^7} = \frac{\binom{8}{1}}{\binom{10}{3}} = \frac{8}{\binom{10}{3}}$$

A coin tossing problem

Assumptions:
- **independence**
- $\mathbf{P}(H) = p$

$$\mathbf{P}\,(k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$

- Given that there were 3 heads in 10 tosses, what is the probability that the first two tosses were heads?
  - event A : the first 2 tosses were heads.
  - event B : 3 out of 10 tosses were heads.
- Second solution: Conditional probability law (on B) is uniform



Length 10 sequence

3 heads sequences

$p^3(1-p)^7$

$$\frac{\#\ in\ (A \cap B)}{\#\ in\ B} = \frac{8}{\binom{10}{3}}$$

Example: The Birthday Problem

- Given there are n Students in a class, find the P (no 2 birthdays coincide)
  - $1 < n < 365$

Ω = all possible sequences

$|\Omega| = (365)^n$

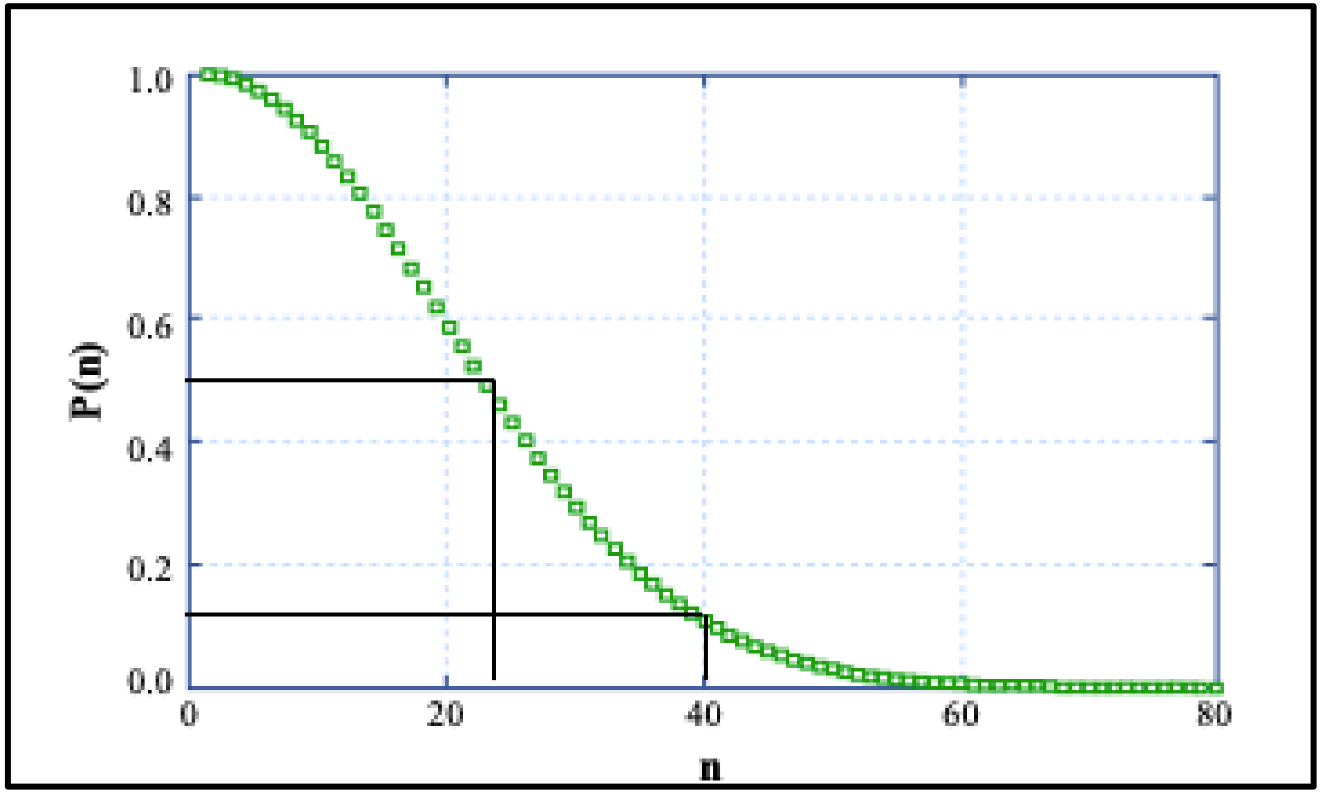
1	2		n^{th} student
365	364	. . . . . .	365−n+1

n students

$$S = 365 * 364 * .... * (365-n+1) = \frac{365!}{(365-n)!}$$

$$P(\text{ no 2 birthdays coincide }) = \frac{365!}{(365-n)!}\frac{1}{(365)^n}$$

Example: The Birthday Problem (ctd)



$$\text{P (} \textit{no 2 birthdays coincide} \text{)} = \frac{365!}{(365-n)!}\frac{1}{(365)^n}$$

Partitions

- $n \geq 1$ distinct items; $r \geq 1$ persons

  Give n_i items to person i

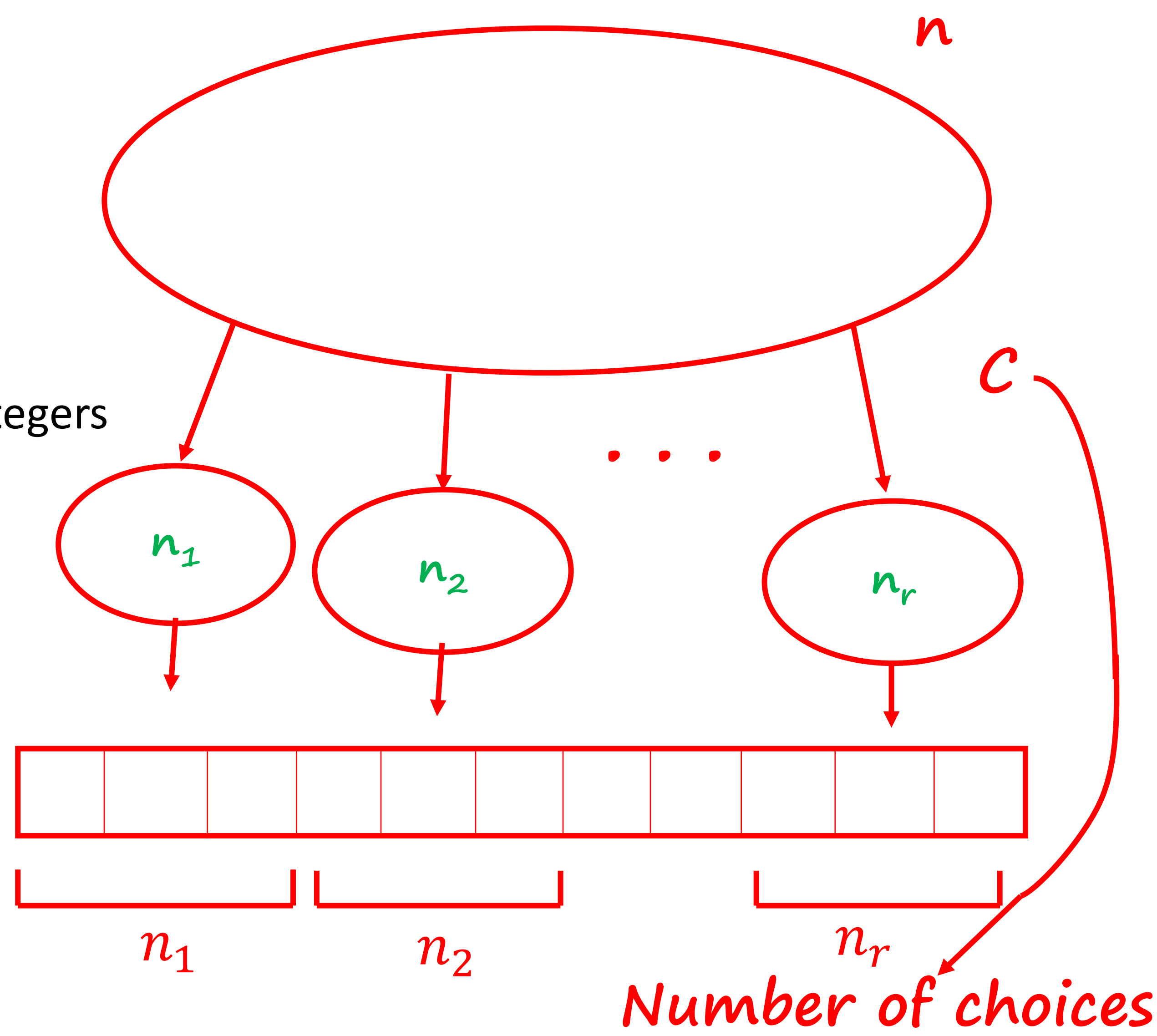
  - Here $n_1, . . , n_r$ are given nonnegative integers
  - With $n_1 + \cdots + \ n_r = n$ are given nonnegative integers

- Ordering n items: $n!$

  - Deal n_i to each person i, and then order

$$C. n_1!\, n_2! \ \ \ldots n_r! = n!$$

Solving for C we get:



$$\text{Number of Choices, } C = \frac{n!}{n_1!\, n_2! \ \ \ldots n_r!} \quad \text{(multinomial coefficient)}$$

Example: 52-card deck, dealt (fairly) to four players.
Find $\mathbf{P}$(each player gets an ace)

- Total possible Number of outcomes: $\dfrac{52!}{13!\,13!\,13!\,13!}$

- Constructing an outcome with one ace to each person: (Distribute the aces) $\dfrac{4!}{1!\,1!\,1!\,1!} = 4 * 3 * 2 * 1$

- Distribute the remaining 48 cards: $\dfrac{48!}{12!\,12!\,12!\,12!}$

- Answer: P(each player gets an ace) $= \dfrac{\dfrac{4!}{1!\,1!\,1!\,1!} * \dfrac{48!}{12!\,12!\,12!\,12!}}{\dfrac{52!}{13!\,13!\,13!\,13!}}$